

AGENTIC AI — CASE STUDY

Placement Assistant

A Coordinator-Routed Multi-Agent Platform for Campus Placement Preparation



PRIMARY CASE STUDY

Placement Assistant — Multi-Agent AI System (2026)

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COMPARATIVE RESEARCH ANCHORS

- LLM Agents in Recruitment — Resume Screening (Gan et al., arXiv 2401.08315, 2024)
- AgentTutor — Multi-Turn Interactive Teaching (Liu et al., AAAI 2026)
- GenMentor — Goal-Oriented Learning ITS (Wang et al., WWW 2025)

The Problem — Why Single-Tool Placement Prep Fails

Context Blur

Resume feedback, coding practice, and interview prep happen in isolation — no shared understanding of student readiness.

No Role Boundaries

A generic chatbot is asked to be resume reviewer, coding mentor, and interviewer at once — instructions blur together.

No Structured Output

Ad-hoc chat replies replace the checklists, scores, and roadmaps students need to act on.

No Progress Memory

Each session starts cold — the system can't tell what topics were mastered last week.

No Personalization

All students get the same generic advice, regardless of target role or current skill level.

The Agentic Fix: Placement Assistant decomposes placement prep into role-specialized agents that exchange structured, standardized artifacts — mirroring how real mentors isolate concerns and hand off verified feedback.

Framework — Defining Properties of Agentic AI

Mapped to This Domain



Role Specialization

Coordinator + 8 agents (Resume, DSA, Aptitude, Company, HR, Technical, Planner, Progress), each with a distinct prompt persona and permitted actions.



Structured Planning

Study Planner turns a goal, timeline, and target company into a day-by-day topic schedule — not free-form chat.



Tool Use

Agents invoke pypdf for resume parsing, Groq LLM API for generation, SQLite for persistence — agents call tools, not just the LLM.



Inter-Agent Communication

A single Coordinator dispatches intent to one specialist function at a time — simpler than a shared message pool.



Autonomy Under a Budget

One student request flows end-to-end to a saved, actionable result (score, roadmap, feedback) with no manual hand-off.

Comparative Analysis I

Resume Agent vs. LLM-Based HR Agent (Gan et al., 2024)

Gan et al. — Resume Screening Pipeline (arXiv 2401.08315)

1

Three Verifiable Stages

Sentence classification → grade & summarize → decide. Removes PII, applies role-based grading, majority-votes final decision.

2

Measured Performance

Fine-tuned LLaMA2-7B-chat reaches F1 87.73% on classification. Processes 838 resumes in 2h55m vs 31h manual — 11× faster.

Placement Assistant — Resume Agent

1

Single-Pass Prompt

Parses PDF → scores ATS-style → lists issues → generates suggestions in one LLM call. Uses Groq-hosted Llama-3.3-70B (not fine-tuned).

2

Gap Identified

No independent F1 or timing evaluation yet. Saves score + feedback to SQLite for Progress dashboard. All three stages in one prompt means accuracy and time-savings are not independently measured.

Gap Analysis: The paper splits resume screening into three verifiable stages with measured F1 and speed-up; the Resume Agent currently does all three in one prompt. This is a natural next step for FA2.

Comparative Analysis II

Learning & Planning Agents vs. AgentTutor & GenMentor

AgentTutor (AAAI 2026)	GenMentor (WWW 2025)	Placement Assistant — Current State
5 modules: Curriculum Decomposition, Assessment, Strategy (LATS tree search), Reflection, Memory. Breaks topics into Bloom's Taxonomy sub-goals. Achieves 92.7% Pass@1 on HumanEval with GPT-3.5-turbo vs 68.1% Reflexion baseline.	Skill-gap identification → adaptive learner profile → personalized path + content. Deployed inside Microsoft's AIEP platform (Oct 2024). Goal-alignment score 4.28/5 with GPT-4o vs 3.51 CoT-prompt baseline.	DSA/Aptitude/Technical Agents use fixed topic roadmaps (Arrays → Strings → ... → DP); not decomposed per-learner by cognitive level. Study Planner generates one-shot day-wise schedule; doesn't re-plan from live performance. Progress Agent stores scores for display only — scores don't feed back into what other agents generate next.

Gap Analysis: Unlike AgentTutor's cognitive decomposition or GenMentor's adaptive profiling, Placement Assistant currently uses static roadmaps and one-shot planning. Implementing feedback loops and adaptive re-planning is a key FA2 improvement.

Comparative Analysis III

Coordination Patterns: Placement Assistant vs. MetaGPT vs. ChatDev

Dimension	Placement Assistant	MetaGPT (ICLR 2024)	ChatDev (ACL 2024)
Coordination Style	Single-dispatch routing	Structured SOP hand-offs	Sequential Chat Chain
# of Agent Roles	9 (1 coordinator + 8)	5 fixed roles	7 fixed roles
Artifact Exchange	Scores, feedback, roadmaps	PRD, UML, API specs, code	Natural language chat log
Self-Correction Loop	Partial (scoring only)	Executable feedback (code runs)	Communicative dehallucination
Best Suited For	End-to-end placement prep	Multi-file software builds	Full SDLC simulation

Reading the Spectrum: Placement Assistant uses the simplest coordination model (single-dispatch), which buys clarity and ease of implementation but sacrifices the richer feedback loops that MetaGPT (executable code) and ChatDev (multi-turn dialogue) employ.

Evidence from Research

Quantitative Benchmarks from Comparative Studies

87.73%

Resume F1 Score

Fine-tuned LLaMA2-7B-chat on sentence classification (Gan et al., 2024)

11×

Faster Screening

2h55m vs 31h manual for 838 resumes (Gan et al., 2024)

92.7%

Pass@1 HumanEval

GPT-3.5-turbo with AgentTutor vs 68.1% Reflexion baseline (AAAI 2026)

4.28/5

Goal-Alignment Score

Skill-gap mapping with GPT-4o (GenMentor, WWW 2025)

⚠ **Caveat:** These numbers benchmark the source frameworks, NOT Placement Assistant. No F1, timing, or Pass@1 evaluation has yet been run on the Resume, DSA, or Technical agents. They are shown here as the standard the research field measures against — a natural next step for FA2.

Honest Assessment — Gaps Versus Published Agentic Frameworks

1

No Closed-Loop Self-Correction

Unlike MetaGPT's executable-feedback loop or AgentTutor's LATS tree search, every agent here produces one LLM pass with no re-run, re-grade, or verification step.

2

No Persistent Skill-Gap Memory

The Progress Agent logs scores to SQLite for display, but — unlike GenMentor's Adaptive Learner Modeling — those scores never feed back into what the DSA, Aptitude, or Planner agents generate next.

3

Company Agent Has No Live Retrieval

Company research comes only from the LLM's general knowledge, with no web search or real-time job board integration — unlike GenMentor's RAG-based Content Curation.

4

One General-Purpose Model for Every Role

All agents use Llama-3.3-70B with role-specific prompts. Unlike Gan et al.'s fine-tuned LLaMA2-7B for resume classification, no agent is fine-tuned on domain-specific data.

5

No Independent Evaluation

The system has not been benchmarked against HumanEval, SWE-Bench, or domain-specific metrics. Accuracy, speed, and user satisfaction are not yet measured.

These gaps are not failures — they are the research frontier. Each one points directly to a concrete FA2 improvement: closed-loop scoring, adaptive re-planning, RAG retrieval, domain fine-tuning, and benchmark evaluation.

References — Foundation of Agentic Frameworks

Gan et al. (2024)

“LLM Agents in Recruitment: Resume Screening...” (arXiv 2401.08315). Demonstrates 87.73% F1 on resume classification and 11× speedup over manual screening.

Liu et al. (AAAI 2026)

“AgentTutor: Agentic Learning with Curriculum Decomposition...” (arXiv 2601.04219). Achieves 92.7% Pass@1 on HumanEval using LATS tree search and multi-turn reflection.

Wang et al. (WWW 2025)

“GenMentor: Goal-Oriented Adaptive Learning...” (arXiv 2501.15749). Deployed in Microsoft's AIEP platform with 4.28/5 goal-alignment score.

AI Policy Disclosure — Transparency in Case Study Analysis

This section details the responsible use of AI tools in preparing this case study analysis.

Literature Retrieval & Fact-Checking

AI tools were used to search for and cross-verify claims against primary sources (Gan et al., AgentTutor, GenMentor, MetaGPT, ChatDev papers).

Summarization Support

AI assisted in condensing lengthy paper sections (architecture, benchmarks, methodology) into concise bullet points while preserving factual accuracy.

Comparative Framework Design

AI helped structure comparative analysis tables and gap assessments for clear, systematic differentiation.

What Was NOT AI-Generated: Case study selection, the honest assessment of gaps, planned extensions, and all specific numerical claims are based on direct reading of primary sources and the Placement Assistant codebase. No synthetic data or fabricated benchmarks were used.

Placement Assistant — System Overview

PRIMARY CASE STUDY

Placement Assistant

ORIGIN

Academic prototype, 2026 | Computer Engineering, Program Elective 4

TECHNOLOGY STACK

Groq API · Llama-3.3-70B · Streamlit · SQLite · Python

ARCHITECTURE

Coordinator Agent + 8 Specialist Agents



Core Claim

A single intelligent platform can personalize, organize, and track placement preparation by routing student requests to specialized agents, each with domain-specific expertise and structured output formats.



Key Innovation

Unlike generic chatbots, the Coordinator Agent reads the student's goal and progress, then routes work to the right specialist — ensuring consistent, role-appropriate feedback and persistent progress tracking across all preparation activities.

System Design — Placement Assistant Architecture

